

A heuristic computation method of the bi-level maximal capacity constrained centroidal power diagram

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The power diagram, a crucial extension of the Voronoi diagram, has been applied across various fields. By integrating capacity and centroid constraints, a capacity constrained centroidal power diagram with preset fixed values could be derived. However, presetting precise capacity values may be challenging, particularly when some sites have fixed locations. To this end, we relax the capacity constraint to an upper bound, termed the maximal capacity constraint. Taking into account both partially location-fixed sites and the maximal capacity constraint, we introduce an improved version of the power diagram called the bi-level maximal capacity constrained centroidal power diagram (Bi-MCCCPD). Then, a heuristic method is proposed to iteratively optimize weights and location-free sites, ultimately generating centroidal power diagrams that adhere to maximal capacity constraints. This involves a weight modification strategy for fine-tuning the weights of sites neighboring these location-fixed sites. Combined with Lloyd's method for optimizing location-free sites, the power diagram with centroid constraints could be produced. Additionally, we use a weight evaluated algorithm to optimize weights, ensuring the power diagrams satisfy the maximal capacity constraints. Experimental results and practical applications validate the effectiveness, feasibility, and practicality of our method.

1. Introduction

The power diagram [1], as an essential advancement of the Voronoi diagram, incorporates weight characteristics for each site, thereby redefining the Euclidean distance as the power distance. By imposing capacity constraints on the ordinary power diagram, the capacity-constrained power diagram (CCPD)

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could be generated. Furthermore, by ensuring that each site aligns with the center-of-mass of its corresponding power cell, the capacity constrained centroidal power diagram (CCCPD) could be derived. The introduction of weight characteristic endows power diagrams with the property of precise capacity constraints. This opens up a wide range of potential applications, including surface remeshing [2, 3, 4], fluid simulation [5, 6], image abstraction [7], blue noise [8, 9, 10], convex decomposition [11], location-allocation problems [12, 13], deep networks [14], etc.

Aurenhammer et al. [1] initially introduce the concepts and properties of the power diagram, also presenting a computational method to generate it from a set of sites. Subsequently, Imai et al. [15] provide a comprehensive overview of the theory and applications of the power diagram. Gavrilova et al. [16] then offer a method to derive the power diagram from the planner Voronoi diagram, albeit with a significant time investment. To enhance the efficiency of constructing the power diagram, Zheng et al. [17] introduced a hybrid GPU-CPU algorithm. This approach utilizes the GPU-based JFA algorithm for rapid construction and rendering of the power diagram within a pixelated domain. Xiao et al. [18] further advanced this by proposing a meshless method. This method involves lifting weighted sites to a higher dimension, combined with the kNN query technique for the power diagram construction. Despite these methods exhibiting high efficiency, they primarily focus on construction strategies rather than optimization designs of the power diagram.

The optimization of the power diagram aims to generate a power diagram that satisfies both capacity and centroid constraints. This involves optimizing weights to meet capacity constraints and adjusting site locations to fulfill centroid requirements. Balzer et al. are the first to present capacity constrained power diagram optimization techniques tailored for discrete [19] and continuous [20] spaces. They combine the false position method, one-by-one iteration, and Lloyd's method to stably compute the CCCPD, but this is time-consuming due to the iterative point-by-point process. de Goes et al. [9] reframe the problem of computing the power diagram with capacity and centroid constraints as a numerical optimization task. They formulate an energy function using the Lagrange multiplier method and employ a combination of Newton's method and an adaptive step-size gradient descent algorithm [21] to achieve faster CCCPD computation. Xin et al. [11] further broaden the scope of the power diagram to a generalized space, proposing an optimization method characterized by super-linear convergence. They adopt the L-BFGS method instead of the adaptive step-size gradient descent technique, enhancing the optimization speed for site locations. Nevertheless, these methods ignore the potential presence of location-fixed sites and the practical challenges of precisely setting capacity values in advance.

Recently, Zheng et al. [22] introduce the interval capacity constraint to power diagram computation, presenting the novel concepts of the variable capacity constrained power diagram (VCCPD) and variable capacity constrained centroidal power diagram (VCCCCPD). However, the introduction of interval capacity constrained values renders the traditional strategy of optimizing weights using Newton's method inapplicable. To address this, they propose a weight evaluation method that iteratively optimizes the weights of variable capacity constrained sites by assessing the equal power distance line varies with weight changes. Meanwhile, Lévy et al. [23], considering partially location-fixed sites, introduce the semi-discrete optimal transport problem and devise a numerical algorithm leveraging a second-order Newton's method. This method is later employed to simulate a universe consisting of over 100 million particles [24]. However, it's worth noting that this method assigns specific values as capacity constraints, posing challenges in practical implementations. In addition, there are few studies simultaneously incorporating variable capacity constraints and partially position-fixed sites in the power diagram computation.

Consider the challenge of pinpointing and assigning shelters during disaster emergencies as a case study. The destruction of partially existing shelters necessitates the building of some new ones with maximal service capabilities to satisfy emergency and evacuation requirements. Inspired by this, we propose an improved version of the power diagram, termed the bi-level maximal capacity constrained power diagram (Bi-MCCPD). This integrates both partial location-fixed sites and maximal capacity constraints simultaneously. In other words, some location-fixed sites are considered in our work, and the capacity of each power cell remains below a preset upper bounded value. Then, we further introduce the bi-level maximal capacity constrained centroidal power diagram (Bi-MCCCCPD) by incorporating the centroid constraint. Leveraging these concepts, we devise a heuristic method that iteratively optimizes weights and location-free sites, ultimately generating a centroidal power diagram with maximal capacity constraints. On the one hand, the weight evaluated algorithm [22] is utilized to optimize the weights, producing power cells that adhere to maximal capacity constraints. On the other hand, a weight modification strategy optimizes the weights of neighboring sites of each location-fixed site based on the relationship between the equal power distance line and weight variation. Combined with Lloyd's method [25] to optimize the positions of location-free sites to the center-of-masses of their power cells, yielding a centroidal power diagram. Ultimately, the Bi-MCCCCPD approach is put to practical application in addressing emergency shelter location challenges, determining the positions and service regions for both existing and upcoming rescue shelters. We contribute the following:

- We present an improved version of the power diagram, dubbed the bi-level maximal capacity constrained centroidal power diagram (Bi-MCCCPD), which takes into account both partially location-fixed sites and maximal capacity constraints concurrently.
- We devise a weight modification strategy that fine-tunes the weight of neighboring sites, generating power cells with centroid constraints for these location-fixed sites.
- We propose a heuristic method for computing the Bi-MCCCPD, integrating a weight evaluated algorithm, Lloyd's method, and the weight modification strategy.

The remainder of this paper is structured as follows. Section 2 briefly reviews the fundamentals of the power diagram and its variants. Section 3 introduces the definitions of the Bi-MCCCPD and the computation methodology. In Section 4, we present our experimental results, while Section 5 explores a practical application of the proposed method. Some conclusions are offered in Section 6.

2. Preliminary

In this section, we briefly introduce the definitions and the power diagram and its variants.

2.1. Power diagrams

The Voronoi diagram [26], also known as the “Voronoi Tessellation”, is a spatial subdivision structure in computational geometry. Given a set of sites (also called “generators”) $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^n$, a d -dimensional domain $\Omega \subset R^d$ is divided into n disjoint subregions $V(\mathbf{X}) = \{V(\mathbf{x}_i)\}_{i=1}^n$. Each subregion (or called “Voronoi cell”) $V(\mathbf{x}_i)$ of the corresponding site $\mathbf{x}_i$ is defined as:

$$(1) \quad V(\mathbf{x}_i) = \{\mathbf{x} \in \Omega | d(\mathbf{x}, \mathbf{x}_i) \leq d(\mathbf{x}, \mathbf{x}_j), \forall j \neq i\}$$

where $d(\mathbf{x}, \mathbf{x}_i) = \|\mathbf{x} - \mathbf{x}_i\|$ is the Euclidean distance between two sites $\mathbf{x}$ and $\mathbf{x}_i$.

The power diagram [19] is an essential extension of the Voronoi diagram, which privileges weight characteristics $\mathbf{W} = \{w_i\}_{i=1}^n$ to sites $\mathbf{X}$. That is, each site $\mathbf{x}_i$ is coincidence with a weight characteristic w_i . Similarly, the power diagram divides the domain Ω into n disjoint subregions $P(\mathbf{X}) = \{P(\mathbf{x}_i)\}_{i=1}^n$, and each subregion (or called “power cell”) $P(\mathbf{x}_i)$ is provided as:

$$(2) \quad P(\mathbf{x}_i) = \{\mathbf{x} \in \Omega | d_p(\mathbf{x}, \mathbf{x}_i) \leq d_p(\mathbf{x}, \mathbf{x}_j), \forall j \neq i\}$$

where $d_p(\mathbf{x}, \mathbf{x}_i) = \|\mathbf{x} - \mathbf{x}_i\|^2 - w_i$ is the power distance between two sites $\mathbf{x}$ and $\mathbf{x}_i$. Notably, the power diagram degenerates to an ordinary Voronoi diagram when the weights of all sites are equal [27, 28].

2.2. Capacity constrained centroidal power diagram

The power diagram has the property of precise capacity constraint owing to the introduction of weight characteristic. By applying the capacity and centroid constraint to an ordinary power diagram, a CCCPD could be obtained [9, 11].

Given that $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^n$ represent n different sites, and each site $\mathbf{x}_i$ is associated with a preset capacity constraint $c_i > 0$. Let $\rho(\mathbf{x})$ denote a C^1 -smooth density function on the domain Ω , the capacity (i.e., area in 2D or volume in 3D) m_i of each power cell $P(\mathbf{x}_i)$ can be calculated as:

$$(3) \quad m_i = \int_{P(\mathbf{x}_i)} \rho(\mathbf{x}) d\mathbf{x}$$

Therefore, the sum of the capacities of power cells is equal to the total capacity of the domain Ω , that is,

$$(4) \quad \sum_{i=1}^n m_i = \int_{\Omega} \rho(\mathbf{x}) d\mathbf{x}$$

Consequently, by optimizing the weight w_i of each site $\mathbf{x}_i$, the power diagram with capacity constraints can be obtained until the capacity m_i is equal to the preset capacity constraint c_i . Furthermore, by imposing the centroid constraint on the power diagram with capacity constraint, the CCCPD can be generated, where each site $\mathbf{x}_i$ is located at the relevant center-of-mass:

$$(5) \quad \mathbf{x}_i = \mathbf{x}_i^* = \frac{\int_{P(\mathbf{x}_i)} \mathbf{x} \rho(\mathbf{x}) d\mathbf{x}}{\int_{P(\mathbf{x}_i)} \rho(\mathbf{x}) d\mathbf{x}}$$

2.3. Variable capacity constrained centroidal power diagram

Different from the precise capacities, Zheng et al. [22] consider the difficulty of presetting an explicit capacity value for each power cell in some practical applications. Based on this, they further relax the capacity constraint from a fixed value to an interval, and then they present a variant of the CCCPD,

called the VCCPD. That is, the capacity m_i of each power cell $P(\mathbf{x}_i)$ lies in a pre-defined interval $[c_i^{min}, c_i^{max}]$:

$$(6) \quad c_i^{min} \leq m_i \leq c_i^{max}$$

Based on this, they further introduce the centroid constraint to the VCCPD, and a VCCCPD could be obtained, where each site is located in the center-of-mass of the corresponding power cell. Consequently, owing to the relaxation of the capacity constraint, VCCPD and VCCCPD are more suitable for these applications with difficulty in presetting exact capacity constraints.

3. Bi-level maximal capacity constrained power diagram

In this section, we introduce the definition of the Bi-MCCCPD, and then we provide a heuristic method to compute the centroidal power diagram with maximal capacity constraints considering both location-fixed and location-free sites.

3.1. Problem description

To elaborate on the fundamental concept of the power diagram, we draw upon the notions introduced in Section 2. Let $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^n$ represent a set of sites in a convex, closed, bounded, and connected d -dimensional domain $\Omega \subset E^d$ (2-dimensional considered in this paper) with a C^1 -smooth density function. $\mathbf{W} = \{w_i\}_{i=1}^n$ denote the associated weight characteristic of sites. According to the definition of the power diagram, the given domain Ω is partitioned into n disjoint subregions $P(\mathbf{X}) = \{P(\mathbf{x}_i)\}_{i=1}^n$, where $P(\mathbf{x}_i) \cap P(\mathbf{x}_j) = \emptyset$, and $\sum_{i=1}^n P(\mathbf{x}_i) = \Omega$.

Without compromising generality, let $\mathbf{X}^\dagger = \{\mathbf{x}_i\}_{i=1}^m$ indicate m ($0 \leq m \leq n$) given m location-fixed sites, and $\mathbf{X}^\uparrow = \{\mathbf{x}_i\}_{i=m+1}^n$ denote the remaining $n - m$ location-free sites. $C^{max} = \{c_i^{max}\}_{i=1}^n$ represent the maximal capacity constraints of given sites $\mathbf{X}$, where $c_i^{max} > 0$ and $\sum_{i=1}^n c_i^{max} \geq \int_{\Omega} \rho(\mathbf{x}) d\mathbf{x}$. That is, the capacity m_i of each power cell $P(\mathbf{x}_i)$, corresponding to the site $\mathbf{x}_i \in \mathbf{X}$, is constrained with an upper bounded value c_i^{max} . Consequently, the Bi-MCCCPD computation aims to iteratively optimize the weights and location-free sites, thereby generating centroidal power diagram with maximal capacity constraints. Similar to [20], the kernel problem of the Bi-MCCCPD

is to minimize the energy function:

$$(7) \quad \begin{aligned} \min \quad & Q(\mathbf{X}^\uparrow, \mathbf{W}) = \sum_{i=1}^n \int_{P(\mathbf{x}_i)} \|\mathbf{x} - \mathbf{x}_i\|^2 \rho(\mathbf{x}) d\mathbf{x} \\ \text{s.t. } & m_i = \int_{P(\mathbf{x}_i)} \rho(\mathbf{x}) d\mathbf{x} \leq c_i^{max}, \forall \mathbf{x}_i \in \mathbf{X} \end{aligned}$$

Due to the coupling of location-fixed sites and maximal capacity constraints, it is challenging to employ existing numerical optimization techniques to minimize the energy function in Equation (7). Therefore, motivated by the optimization strategy in [22], we propose a heuristic computation method to generate centroidal power diagram with maximal capacity constraints considering both location-fixed and location-free sites.

3.2. Heuristic computation method

In the following, we describe the heuristics method for generating the Bi-MCCCCPD.

3.2.1. Overall idea Inspired by previous work [9, 11, 22] on power diagram optimization, weight optimization and site optimization are performed separately in the Bi-MCCCCPD iterative optimization process. Besides, the site optimization strategies for different location-constrained types of sites vary due to the existence of both location-fixed and location-free sites. Specifically, the key operations in our heuristic method are summarized: (1) optimizing the weights of sites to satisfy the maximal capacity constraint; (2) moving these location-free sites to the center-of-mass of the corresponding power cells by Lloyd's method; and (3) modifying the weights of these neighboring sites of location-fixed sites to meet the centroid constraint. These three steps are executed iteratively and alternatively until the maximal capacity constraint and the centroid constraint are satisfied. The pseudo-code of the proposed heuristic method for computing the Bi-MCCCCPD is presented in Algorithm 1.

3.2.2. Weight optimization Previous research has demonstrated the efficacy of various numerical techniques for accelerating weight optimization, including the adaptive step-size gradient descent method [21], Newton's method [9], and quasi-Newton method [11], among others. However, these techniques have been primarily applied to power diagram computation with determined capacity constrained values. The relaxation of capacity constraints to maxi-

Algorithm 1 Heuristic algorithm for the Bi-MCCCPD

Input: Domain Ω , number of sites n , location-fixed sites $\mathbf{X}^\downarrow = \{\mathbf{x}\}_{i=1}^m$, maximal capacity constraints $\mathbf{C}^{max}$, centroid distance termination ϵ

Output: Power diagram $(\mathbf{X}, \mathbf{W})$

- 1: Randomly initialize $n - m$ sites $\mathbf{X}^\uparrow$ as location-free sites
- 2: Merge all sites $\mathbf{X} = \{\mathbf{X}^\downarrow, \mathbf{X}^\uparrow\}$, and set the weights $\mathbf{W} = \mathbf{0}$
- 3: **while** $\exists \mathbf{x}_i, m_i > c_i^{max}$ or $\|\mathbf{x}_i - \mathbf{x}_i^*\| > \epsilon$ **do**
- 4: **for** $\mathbf{x}_i \in \mathbf{X}$ **do**
- 5: **if** $m_i > c_i^{max}$ **then**
- 6: Compute the nearest distance d_i
- 7: Calculate the weight variation $\Delta w_i = 2d_i \cdot \Delta l_{ij}$
- 8: Update weight $w_i = w_i - \Delta w_i$
- 9: **end if**
- 10: **end for**
- 11: Update power diagram based on modified weights
- 12: **for** $\mathbf{x}_i \in \mathbf{X}$ **do**
- 13: Compute the center-of-mass $\mathbf{x}_i^*$ of $P(\mathbf{x}_i)$
- 14: **if** $\mathbf{x}_i \in \mathbf{X}^\uparrow$ **then**
- 15: Update location $\mathbf{x}_i \leftarrow \mathbf{x}_i^*$
- 16: **else**
- 17: Search the neighboring sites $\mathbf{N}_{\mathbf{x}_i}$ of $\mathbf{x}_i$
- 18: **for** $\mathbf{x}_j \in \mathbf{N}_{\mathbf{x}_i}$ **do**
- 19: Calculate the weight variation Δw_j
- 20: Update weight $w_j = w_j + \text{sgn}(\overrightarrow{\mathbf{x}_i \mathbf{x}_j^*}, \overrightarrow{\mathbf{x}_i \mathbf{x}_j}) \cdot \Delta w_j$
- 21: **end for**
- 22: **end if**
- 23: **end for**
- 24: Update power diagram based on sites and weights
- 25: **end while**

mal values in the Bi-MCCCPD presents a challenge for the direct application of these numerical techniques. With this regard, the weight evaluated algorithm in [22] is utilized to optimize the weights of sites, thereby generating a power diagram with maximal capacity constraints, as shown in Figure 1.

Specifically, as shown in Figure 1, let $\mathbf{x}_{ij}$ (or $\mathbf{x}'_{ij}$ after weight changed) denote the intersecting point of $\mathbf{x}_i \mathbf{x}_j$ and the equal power distance line, and l_{ij} (or l'_{ij} after weight changed) represent the distance between $\mathbf{x}_i$ and $\mathbf{x}_{ij}$, i.e., $l_{ij} = \|\mathbf{x}_{ij} - \mathbf{x}_i\|$. Based on the power diagram definition [1, 9], the equal power distance line variation $\Delta l_{ij} = l'_{ij} - l_{ij}$ is calculated:

$$(8) \quad \Delta l_{ij} = \left\| \frac{w'_i - w_i}{2\|\mathbf{x}_j - \mathbf{x}_i\|^2} \cdot (\mathbf{x}_j - \mathbf{x}_i) \right\| = \frac{\Delta w_i}{2\|\mathbf{x}_j - \mathbf{x}_i\|}$$

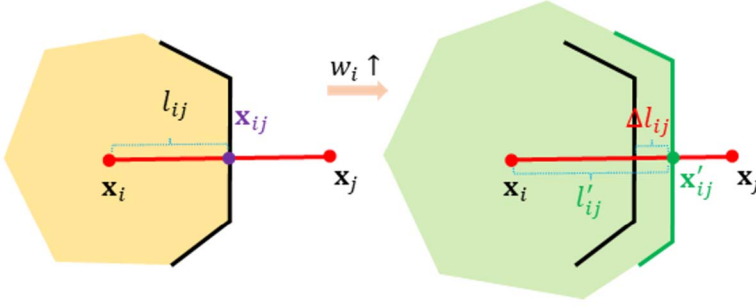


Figure 1: The equal power distance line variation Δl_{ij} between two adjacent sites $\mathbf{x}_i$ and $\mathbf{x}_j$ with the weight w_i increases, where the green line indicates the equal power distance lines with weight changed.

Therefore, the calculation of weight variation Δw_i can be formulated as $\Delta w_i = 2d_{ij} \cdot \Delta l_{ij}$, where d_{ij} represents the distance between site $\mathbf{x}_i$ and its adjacent facilities. However, a challenge arises due to the variance in these distances d_{ij} , which complicates the determination of weight variations Δw_i . To circumvent this issue, we adopt the approach utilized in [3], where the baseline for computing the weight variations is built on the distance d_i between site $\mathbf{x}_i$ and its nearest neighboring site. This strategy not only enables an effective circumvent of potential empty power cells during the optimization process but also ensures a robust and practical solution. Besides, the value of Δl_{ij} is set to 0.05, that could be referred [22] in detail. Consequently, the pseudo-code for weight optimization to satisfy the maximal capacity constraint is presented in Algorithm 1 (lines 4 to 10).

3.2.3. Site optimization In terms of site optimization in the power diagram, Lloyd's method [25] and the quasi-Newton method (e.g., the L-BFGS method) [11] are the most frequently employed techniques. In our work, taking these location-free sites into account, Lloyd's method is utilized to optimize their locations. In other words, each location-free site $\mathbf{x}_i \in \mathbf{X}^\uparrow$ is relocated to the center-of-mass $\mathbf{x}_i^*$ of its corresponding power cell $P(\mathbf{x}_i)$, as specified by Equation (5), thereby satisfying the centroid constraint of these location-free sites.

However, the consideration of partially location-fixed sites leads to the challenge of developing a particular technique to optimize the power cell of these location-fixed sites. One prevalent idea is to optimize the weights of these location-fixed sites. Nevertheless, it could be observed that variations in the equal power distance lines of different adjacent sites are heterogeneous

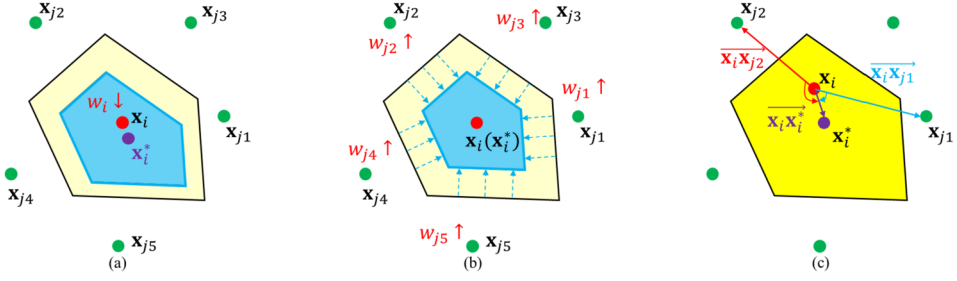


Figure 2: The weight modification strategy for these location-fixed sites in the Bi-MCCCPD computation process. (a) the power cell (shown in the blue polygon) is optimized by reducing the weight w_i , but this may causes $\mathbf{x}_i$ to be potentially out of the center-of-mass $\mathbf{x}_i^*$ of the optimized power cell (shown in the purple dot); (b) the power cell is optimized by increasing the weights of these neighboring sites independently to preserve the centroid constraint of $\mathbf{x}_i$; and (c) the weight modification strategy for these location-fixed sites with centroid constraint.

due to changes in the weight w_i of site $\mathbf{x}_i$, as illustrated in Figure 2(a). This heterogeneity significantly disrupts the centroid constraint for location-fixed sites. To this end, we propose a heuristic technique, called the weight modification strategy, to achieve the centroid constraint of these location-fixed sites. This strategy updates the weights of its adjacent sites $\mathbf{N}_{x_i}$ based on the relationship between the equal power distance line and the weight variation, instead of the weight of each location-fixed site, as shown in Figure 2(b–c). To be specific, for each adjacent site $\mathbf{x}_j \in \mathbf{N}_{x_i}$, we calculate the dot product $\text{sgn}(\overrightarrow{\mathbf{x}_i \mathbf{x}_i^*}, \overrightarrow{\mathbf{x}_i \mathbf{x}_j})$ between two vectors, $\overrightarrow{\mathbf{x}_i \mathbf{x}_i^*}$ and $\overrightarrow{\mathbf{x}_i \mathbf{x}_j}$, to facilitate the weight w_j update process as follows:

- $\text{sgn}(\overrightarrow{\mathbf{x}_i \mathbf{x}_i^*}, \overrightarrow{\mathbf{x}_i \mathbf{x}_j}) > 0$ (acute angle), the weight w_j is increased by Δw , as shown by $\mathbf{x}_{j1}$ in Figure 2(c);
- $\text{sgn}(\overrightarrow{\mathbf{x}_i \mathbf{x}_i^*}, \overrightarrow{\mathbf{x}_i \mathbf{x}_j}) < 0$ (obtuse angle), the weight w_j is decreased by Δw , as shown by $\mathbf{x}_{j2}$ in Figure 2(c);
- $\text{sgn}(\overrightarrow{\mathbf{x}_i \mathbf{x}_i^*}, \overrightarrow{\mathbf{x}_i \mathbf{x}_j}) = 0$ (right angle), the weight w_j should not be updated;

where the value Δw could be calculated based on Equation (8).

By integrating Lloyd's method with the weight modification strategy, we can optimize the location of each site to coincide with the center-of-mass of its corresponding power cell. The pseudo-code for the site optimization process is presented in Algorithm 1 (lines 12 to 23).

4. Experiments

In this section, we demonstrate a comprehensive experimental results to verify the effectiveness and capability of our method. All experiments are conducted on a Windows 11 computer with an Intel (R) Core (TM) i9-13900H CPU operating at 2.6 GHz and equipped with 32.0 GB of memory. The default parameter setting in our work is as follows: the percentage $\gamma = m/n$ of location-fixed sites is set to 10%, and the default maximal capacity constraint of each site is $1.1\bar{m}$, where $\bar{m} = \frac{1}{n} \cdot \int_{\Omega} \rho(\mathbf{x}) d\mathbf{x}$. The equal power distance line variation Δl_{ij} in the optimization process is set to 0.05 in our experiment.

4.1. Results analysis

To validate the effectiveness of the proposed method, we first conduct several experiments on various domains, capacity constraints, and density functions.

4.1.1. Complex capacity constraint analysis Maximal capacity constraints and considerations for both location-fixed and location-free sites constitute essential components of our work. To assess the feasibility of the proposed method in computing the Bi-MCCCPCD, we initially showcase the optimized results on squared and non-convex domains, incorporating a range of complex capacity constraints. Specifically, the squared and non-convex urban domains with uniform density serve as baselines, with the percentage γ of location-fixed sites set as 10%. Additionally, both the default and differential maximal capacity constraints are taken into account in this experiment. Figure 3 illustrates the computational results, including optimized location-free sites and power cells, for 50 and 100 sites. The red pentagrams denote these location-fixed sites, while polygons colored in yellow represent sites with differential capacity constraints. Notably, the varying shades of yellow in Figure 3 correspond to different levels of maximal capacity constraints, as depicted in Figure 3(c) and Figure 3(f).

The results presented in Figure 3 demonstrate the efficacy of the proposed method in computing centroidal power diagram with maximal capacity constraint, combining the location-fixed and location-free sites. Given the configurations of domain shapes, our method efficiently pinpoints the locations and power cells. The location of each site is optimized to the center-of-mass of its corresponding power cell, achieving the centroid constraint. Considering the differential capacity constraints, the polygons shaded with varying degrees of yellow illustrate the optimized results of pertinent sites, further underscoring the efficacy of the proposed method.

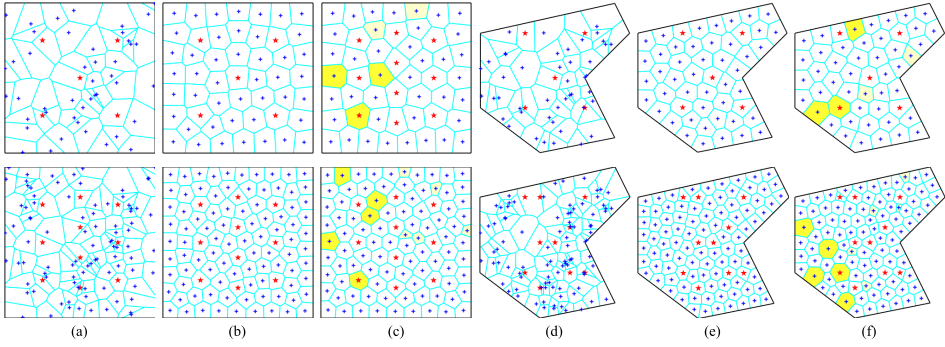


Figure 3: The computational results of 50 and 100 sites on squared and non-convex domains with uniform density function. The red pentagrams and blue plus-signs indicate these location-fixed and location-free sites, respectively. The polygons formed by cyan boundaries show the optimized power cells to the corresponding sites. Top: results with 50 sites; bottom: results with 100 sites. (a)–(c) the initialized results, optimized results with the same maximal capacity constraints, and optimized results with differential maximal capacity constraints (shown in yellow polygons) in a squared domain. (d)–(f) the initialized results, optimized results with the same maximal capacity constraints, and optimized results with differential maximal capacity constraints (shown in yellow polygons) in a non-convex polygonal domain.

4.1.2. Density analysis Typically, using uniform density to characterize density distributions is inappropriate in practical applications. Previous studies have employed various continuous density functions in terms of specific requirements, such as linear density (LD), and non-linear density (NLD), etc. With this regard, we conduct an experiment to assess the reliability of the proposed method under various densities, as shown in Figure 4. In this experiment, two linear densities (LD-1 and LD-2) and two non-linear densities (NLD-1 and NLD-2) are taken into consideration. The maximal capacity constraint for each site is set to the default value, and the percentage γ of location-fixed sites is also set to 10%. The computational results presented in Figure 4 illustrate the optimized sites and power cells in a combination of 2×2 density functions.

These results in Figure 4 highlight the dependability of the proposed method in computing centroidal power diagram with maximal capacity constraint, considering both location-fixed and location-free sites. The optimized power cells and sites in these results align with the domain density distribution. In essence, dense regions are assigned more sites with smaller power

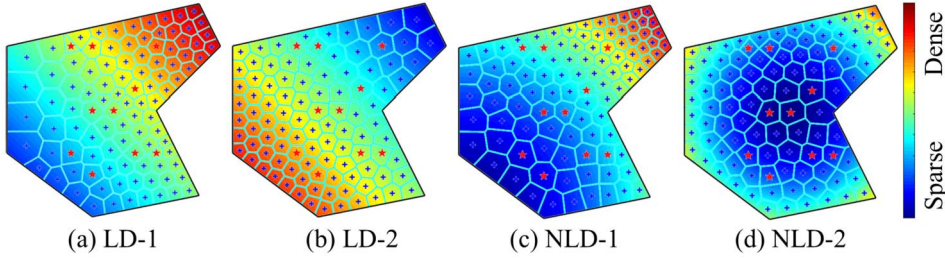


Figure 4: Computational results of 100 sites with various density functions. (a) result with linear density function (LD-1): $\rho(\mathbf{s}) = 0.1 + x + y$, (b) result with linear density function (LD-2): $\rho(\mathbf{s}) = 2.1 - x - y$, (c) result with non-linear density function (NLD-1): $\rho(\mathbf{s}) = 0.1 + x^2 + y^2$, and (d) result with non-linear density function (NLD-2): $\rho(\mathbf{s}) = 0.1 + 2 \times ((x - 0.5)^2 + (y - 0.5)^2)$.

cells, while fewer sites with large power cells are located in sparse regions, thus satisfying maximal capacity constraints.

4.2. Performance analysis

To verify the effectiveness of the proposed method, we conduct an experimental analysis of the computational efficiency under various combinations of densities and percentages of fixed-sites. Moreover, to corroborate the efficacy and superiority of our method, a comparative evaluation is performed against existing methods, including CCCPD [11], and VCCCCPD [22]. This rigorous comparison enables a thorough assessment of the relative performance of our method against benchmarks.

4.2.1. Computational efficiency The percentage γ of location-fixed sites plays an essential role in the proposed method, which affects the optimized power diagrams. Therefore, we conduct an experiment to clarify the computational results of the proposed method in a uniformly distributed domain with varying percentages γ of location-fixed sites, as shown in Figure 5. Besides, we also count the computational time of the proposed method in uniform (UD), linear (LD-1), and non-linear (NLD-2) domains for 50 and 100 sites, as shown in Figure 6. Note that, the maximal capacity constraints are set to the default values in this experiment.

According to the results presented in Figure 5 and Figure 6, it is evident that the number of location-fixed sites has a direct impact on the optimized power diagrams. Furthermore, this factor γ also significantly affects the computational efficiency of the proposed method. From a qualitative perspective,

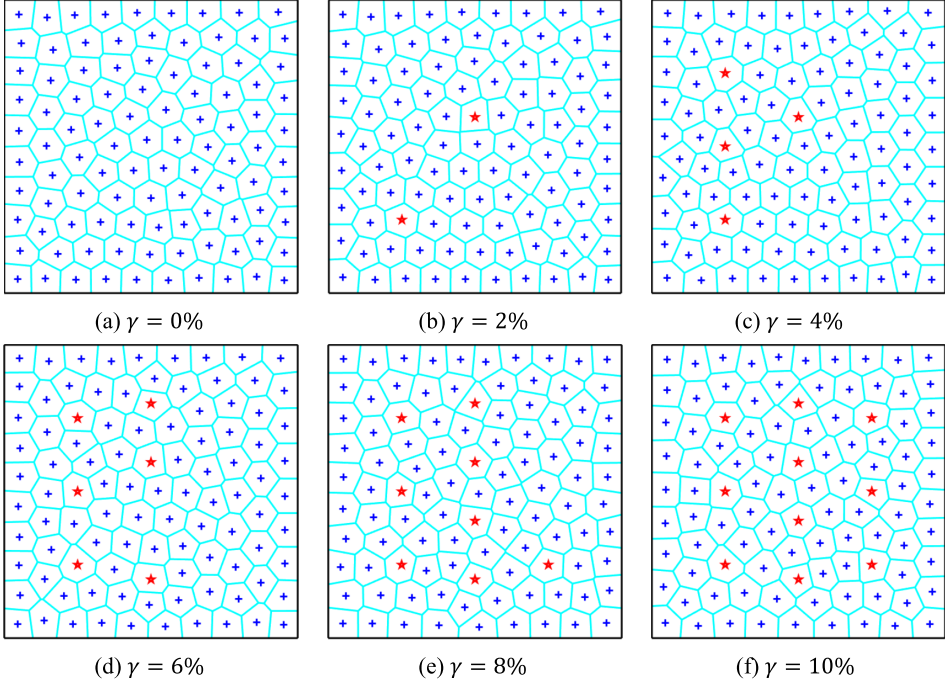


Figure 5: The computational results of the proposed method with different percentages γ of location-fixed sites. These red pentagrams and blue plus-signs indicate the location-fixed and location-free sites, respectively.

as illustrated by the red pentagrams in Figure 5, several location-fixed sites are presented in the given domain. This restriction on the selection of too-close locations for location-free sites (depicted as blue plus-signs in Figure 5) affects their location and weight optimization for neighbouring location-free sites. Quantitatively, as shown in Figure 6, the number of location-fixed sites plays a crucial role in the computational complexity and overall optimization process. The results indicate that an increase in the number of location-fixed sites significantly increases the computational time and reduces the efficiency of the proposed method.

4.2.2. Comparison To further corroborate the efficacy and superiority of the proposed method, a comparative analysis is conducted against several representative power diagram optimization methods, including the false-position method [20], the super-linear method [11] and the VCCPD [22]. Notably, the power diagram generated by the proposed method degenerates to a CC-CPD when the exponential parameter α is 1.0. Despite numerous efficient

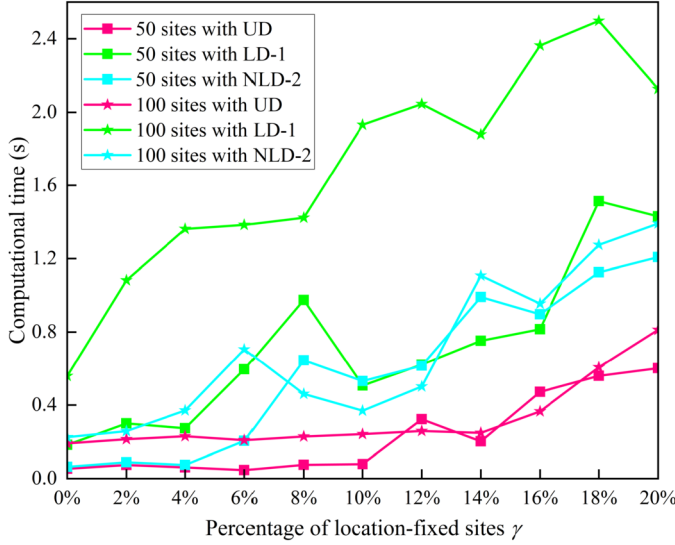


Figure 6: Computational efficiency of the proposed method with different percentages γ of location-fixed sites.

algorithms [29] have been put forth to expedite the optimization of site locations, our primary focus lies exclusively on the maximal capacity constraints and location-fixed sites. Therefore, these methods used for comparison in this experiment are re-implemented by Lloyd's method for location optimization. Besides, Newton's method [9] and the weight evaluated algorithm [22] are employed to optimize weights in the super-linear and VCCCPCD frameworks, respectively. In the false-position method and the super-linear method, the fixed capacity constraint is set to the average value $\overline{m}_i$, while the VCCCPCD utilizes a pre-defined capacity constraint interval of $[0.9 \cdot \overline{m}_i, 1.1 \cdot \overline{m}_i]$. The results of our comparative analysis are comprehensively presented in Figure 7, which illustrates the optimized power diagram for 50 sites using various methods. The corresponding qualitative and quantitative comparison results are provided in Table 1. This experimental evaluation considers three densities: uniform (UD), linear (LD-1), and non-linear (NLD-1), as previously shown in Figure 4. Moreover, a combination of two optimized power diagrams are examined by each method: one involving only location-free sites (called "No-fixed"), and the other incorporating a combination of 10% location-fixed and 90% location-free sites (namely "Hybrid"). In these visualizations, location-free sites are represented by blue plus-signs, while location-fixed sites are denoted by red pentagrams.

Table 1: Qualitative and quantitative comparison results of the false-position method [20], the super-linear method [11], the VCCCPD [22], and our method, corresponding to these results in Figure 7

Methods	Capacity	Site	Computational time (s)		
			UD	LD-1	NLD-1
Balzer et al. [20]	Fixed	No-fixed	1.075	1.527	1.683
		Hybrid	1.736	1.945	2.318
Xin et al. [11]	Fixed	No-fixed	0.787	0.968	0.997
		Hybrid	1.341	1.420	1.589
Zheng et al. [22]	Interval	No-fixed	0.090	0.180	0.251
		Hybrid	0.099	0.246	0.304
Ours	Maximal	No-fixed	0.046	0.156	0.140
		Hybrid	0.396	0.426	0.410

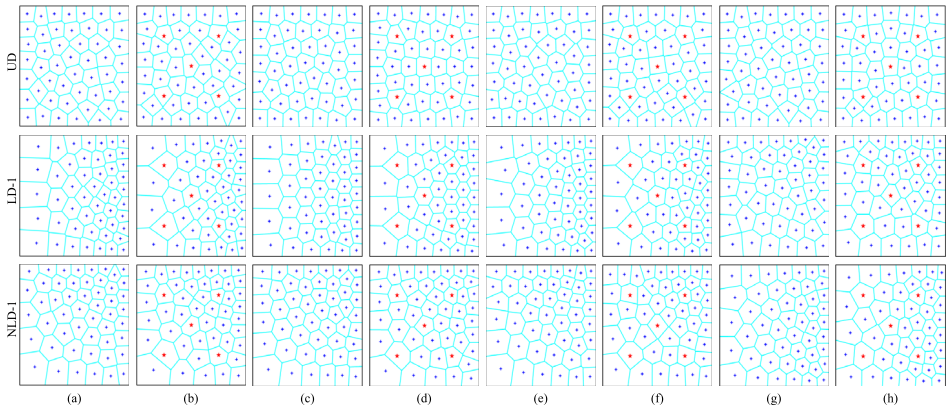


Figure 7: Comparison results of the power diagrams with various methods, including the false-position [20], the super-linear [11], the VCCCPD [22], and the proposed method. From top to bottom: results of various methods in uniform (UD), linear (LD-1), and non-linear (NLD-1) densities. From left to right: (a) results generated by the false-position method [20] without location-fixed sites; (b) results generated by the false-position method [20] with 10% location-fixed sites; (c) results generated by the super-linear method [11] without location-fixed sites; (d) results generated by the super-linear method [11] with 10% location-fixed sites; (e) results generated by the VCCCPD [22] method without location-fixed sites; (f) results generated by the VCCCPD [22] method with 10% location-fixed sites; (g) results generated by the proposed method without location-fixed sites; and (h) results generated by the proposed method with 10% location-fixed sites.

The results shown in Figure 7 and Table 1 demonstrate the efficacy and superiority of the proposed method when compared to other existing approaches. Specifically, the false-position method attempts to optimize the weights by iteratively modifying the weight for each site, but this is inefficient for precise capacity constrained power diagram computation. The super-linear method [11] applies Newton’s method to optimize the weights, coupled with Lloyd’s method to optimize the sites, achieving the generation of centroidal power diagram with capacity constraints. However, this often requires more iterations, which translate to increased time consumption, due to the fixed-value capacity constraints. Different from that, the VCCCCPD method [22] relaxes the capacity constraint from a fixed-value to an interval. Based on this, the weight evaluated algorithm is taken to optimize the weights, generating power diagram satisfied interval capacity constraints. Nevertheless, a common limitation shared by these methods is the difficulty in optimizing location-fixed sites, as shown in Figure 7(b), Figure 7(d) and Figure 7(f).

Fortunately, the proposed method addresses the limitation in previous work by integrating both the maximal capacity constraint and the consideration of location-fixed sites. That is, we ensure that capacities of sites do not exceed the pre-defined maximum values. Besides, the location-fixed sites are taken into consideration, and the capacities of the corresponding power cells are optimized based on the weights of adjacent sites. This integrated method offers a more comprehensive and effective solution for partial location-fixed applications, such as location-allocation problems, etc.

4.3. More results

To further assess the effectiveness of the proposed method, we investigate the computational results for a wider range of site scenarios. In this experiment, we compute and record the results for three combinations consisting of various sites and densities. To be specific, we utilize the domains of uniform and linear densities, with the total number of sites set to 200, 500, and 1000, respectively. The percentage γ of location-fixed sites is set to 5%. The computational results of power diagrams are demonstrated in Figure 8, where plus-signs indicate these initialized location-free sites, and pentagrams represent these location-fixed sites. The results in Figure 8 prove that the proposed method could generate Bi-MCCCCPD with more sites.

5. Applications

To demonstrate the feasibility and utility of the proposed Bi-MCCCCPD computation method, we further conduct an experiment in a representative unitized urban domain. Specifically, eight parks or campuses are identified as

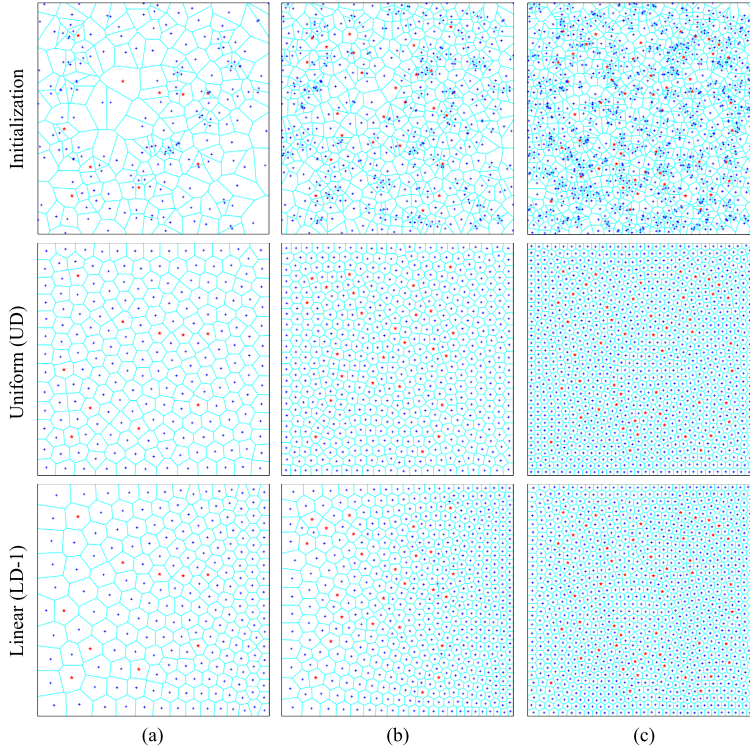


Figure 8: Computational results of the proposed method with more sites. From top to bottom: initial power diagram, optimized results with uniform density, and optimized results with linear (LD-1) density. From left to right: (a) 200 sites; (b) 500 sites; and (c) 1000 sites.

potential emergency rescue facilities within the urban domain, as shown in Figure 9(a). Assuming an urban disaster, the population would require evacuation to these emergency rescue facilities. However, due to scale and capacity constraints, the established parks and campuses cannot accommodate all individuals. Therefore, temporary emergency rescue facilities must be established, necessitating location selection and service region allocation.

In this experiment, a total of 100 emergency rescue facilities, comprising 92 pending temporary facilities, are strategically located in the urban domain to cater to the requirements of all personnel. Taking into account the distinct population distribution characteristics in the urban domain, two distinct density models are employed to approximate the population distributions at various stages. Firstly, the single-center population distribution

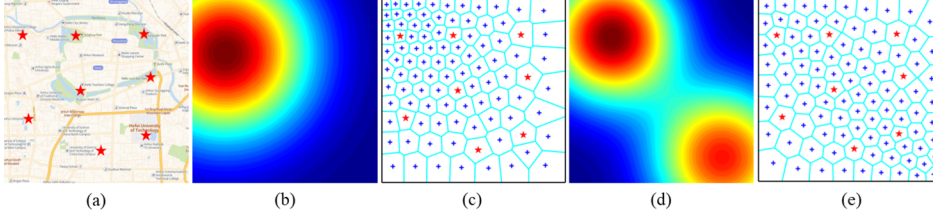


Figure 9: The locations and service region allocations of 100 emergency rescue facilities in a certain city, where eight parks and campuses are utilized as the established facilities. (a) an urban domain with 9 established emergency rescue facilities, (b) a single-center population distribution $\rho(\mathbf{s}) = 5.858 \times e^{-5 \times [(x-0.23)^2 + (y-0.79)^2]} - 0.005 \times [(x-0.23)^2 + (y-0.79)^2]^{0.5}$, (c) the location-allocation result of 100 emergency rescue facilities with the single-center population distribution, (d) a multi-center population distribution $\rho(\mathbf{s}) = 3.271 \times e^{-8.25 \times [(x-0.23)^2 + (y-0.79)^2]} + 2.393 \times e^{-7.096 \times [(x-0.83)^2 + (y-0.14)^2]}$, and (e) the location-allocation result of 100 emergency rescue facilities with the multi-center population distribution.

model, as depicted in Figure 9(b), highlights a monocentric distribution of population in the urban domain, with the population gradually decreasing from the center outwards. Secondly, the multi-center population distribution model, shown in Figure 9(d), accounts for two centers of population concentration in the urban domain. These models allow for a comprehensive analysis and understanding of population distribution and its impact on the location and allocation of emergency rescue facilities within the urban domain. Figure 9(c) and Figure 9(e) present the locations and allocation service regions of 100 emergency rescue facilities in a single-center and a multi-center distributed population in the urban domain, respectively. Notably, the maximal capacity constraint of each emergency rescue facility is set to the default value in this experiment. From the results in Figure 9, we could observe that the proposed is feasible to solve the bi-level location-allocation problem in practical applications. The allocated service region of each emergency rescue facility is no more than the predefined maximal capacity constraint, that coincides with the actual urban disaster scenarios.

6. Conclusion

This paper introduces an extension of the power diagram, called Bi-MCCCCPD, which considers both partial location-fixed sites and maximal capacity constraints. Based on this, a heuristic method is proposed to iteratively and

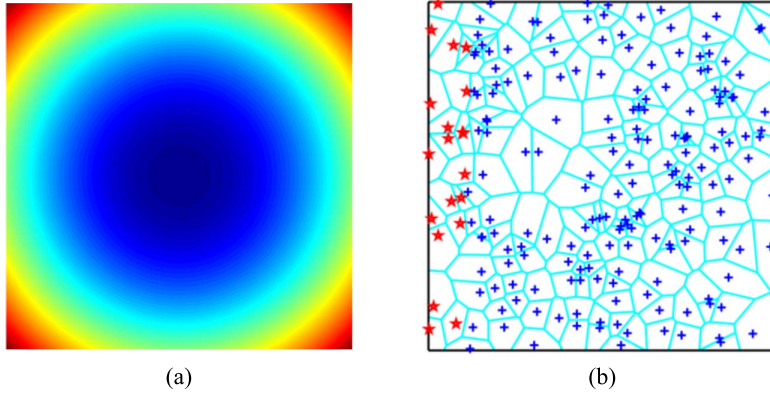


Figure 10: An example of failure with too many fixed sites in a sparse-density region. (a) density map; (b) initialization with too many fixed sites in a sparse-density region.

alternatively optimize the weight and location-free sites. This method combines the weight evaluated algorithm for weight optimization, Lloyd’s method for location-free site optimization, and a weight modification strategy for location-fixed site optimization, thereby achieving the generation of centroidal power diagrams with maximal capacity constraints. Furthermore, the Bi-MCCCPD is utilized and evaluated to solve the locating problem of emergency shelters in practical applications. The experimental results demonstrate the effectiveness, feasibility, and usefulness of the proposed method.

Limitations and future work There are still a few limitations in our work. One is that these location-fixed sites have an impact on the optimization of the proposed heuristic method. That is, our method may generate incorrect results or even fail in sparsely density regions with too many location-fixed sites, as shown in Figure 10. This could be addressed by applying additional constraints for location-fixed sites. The other is that the variant of the weight evaluation method for these location-fixed sites is inefficient with overly many sites. Consideration may be given to a parallel computing technique [3] or dynamic parameter adjustment mechanism [20], etc.

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